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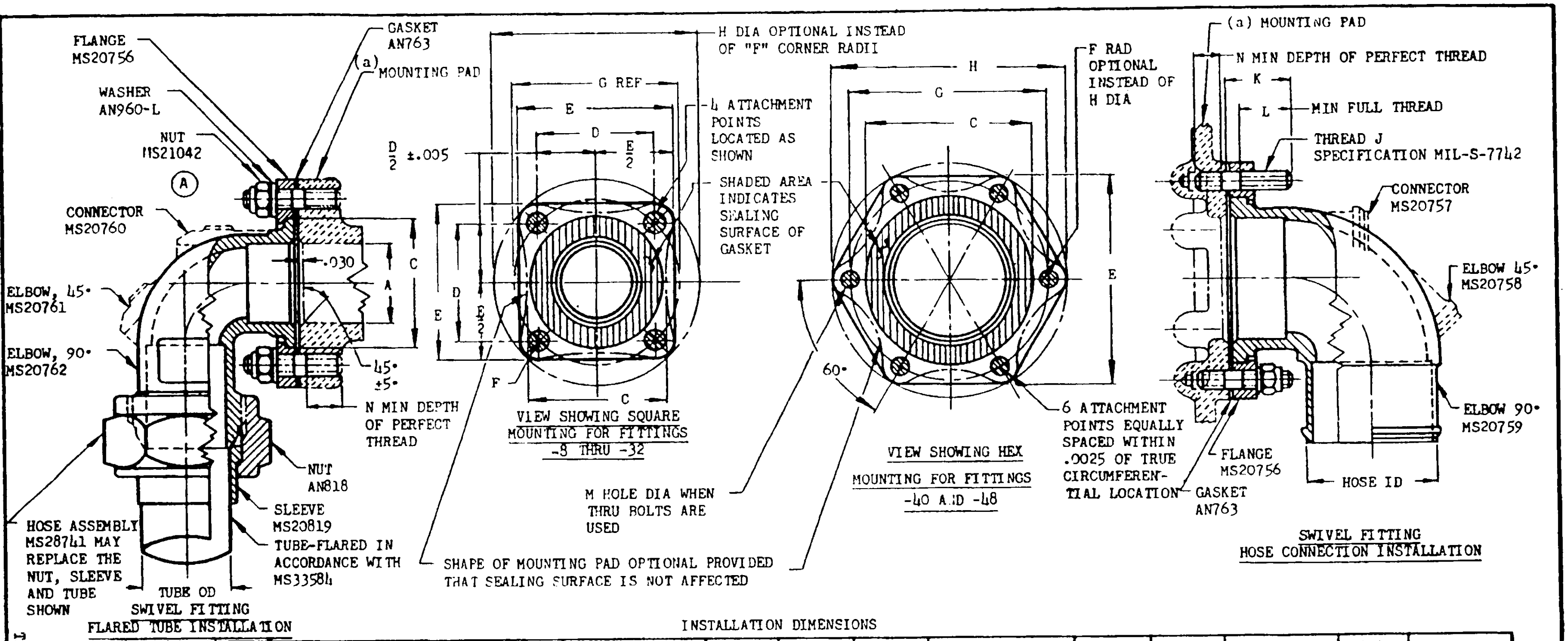
Review activities: DLA- CS

User activities: ARMY - ME


FLANGE MS20756
WASHER AN960-L
NUT MS21042
Ⓐ
CONNECTOR MS20760
ELBOW, 45° MS20761
ELBOW, 90° MS20762
GASKET AN763
(a) MOUNTING PAD
D/2 ±.005
.030
C
A
45° ±5°
N MIN DEPTH OF PERFECT THREAD
NUT AN818
SLEEVE MS20819
TUBE-FLARED IN ACCORDANCE WITH MS33584
HOSE ASSEMBLY MS28741 MAY REPLACE THE NUT, SLEEVE AND TUBE SHOWN
TUBE OD
G REF
E
D
E/2
F
C
H DIA OPTIONAL INSTEAD OF "F" CORNER RADII
4 ATTACHMENT POINTS LOCATED AS SHOWN
SHADED AREA INDICATES SEALING SURFACE OF GASKET
M HOLE DIA WHEN THRU BOLTS ARE USED
SHAPE OF MOUNTING PAD OPTIONAL PROVIDED THAT SEALING SURFACE IS NOT AFFECTED
H
G
C
E
60°
F RAD OPTIONAL INSTEAD OF H DIA
6 ATTACHMENT POINTS EQUALLY SPACED WITHIN .0025 OF TRUE CIRCUMFERENTIAL LOCATION
(a) MOUNTING PAD
N MIN DEPTH OF PERFECT THREAD
K
L
MIN FULL THREAD
THREAD J SPECIFICATION MIL-S-7742
CONNECTOR MS20757
ELBOW 45° MS20758
ELBOW 90° MS20759
FLANGE MS20756
GASKET AN763
HOSE ID

SWIVEL FITTING
FLARED TUBE INSTALLATION

VIEW SHOWING SQUARE
MOUNTING FOR FITTINGS
-8 THRU -32

VIEW SHOWING HEX
MOUNTING FOR FITTINGS
-40 AND -48

SWIVEL FITTING
HOSE CONNECTION INSTALLATION

INSTALLATION DIMENSIONS

DASH NO. REF	NOMINAL TUBE AND HOSE SIZE REF	A ±.003 DIA	C +.000 -.005 DIA	D ±.005	E		F RAD	G ±.005 DIA	H DIA	THREAD J	K ±.020	L	M BOLT HOLE +.010 -.000	(c) N	(b) WRENCH TORQUE INCH LBS.
-8	.500	.383	.875	.950	1.390	±.016	.219	1.344	1.782	NO. 10-32 NF-3A	.625	.500	.205	.281	40
-10	.625	.469	1.000	1.038	1.468			1.468	1.906						
-12	.750	.609	1.250	1.156	1.594			1.635	2.094						
-12-16			1.500	1.312	1.750			1.855	2.312						
-16	1.000	.844													
-20	1.250	1.078	1.844	1.656	2.188	±.020	.266	2.342	2.875	1/4 -28 UNF-3A	.688	.562	.266	.375	100
-20-24			2.125	1.812	2.375		.281	2.562	3.094		.750	.625			
-24	1.500	1.312													
-32	2.000	1.781	2.750	2.375	3.000		.312	3.359	3.953	5/16-24 UNF-3A	.812	.688	.328	.469	225
-40	2.500	2.281	3.281		4.000		.375	3.812	4.500						
-48	3.000	2.781	3.781		4.500		.625	4.312	5.000						

(a) DETAIL DESIGN AND CONSTRUCTION OF SWIVEL FITTING MOUNTING PAD OPTIONAL EXCEPT THAT GASKET SEALING SURFACE, AS NOTED, MUST BE FINISHED SMOOTH WITH A SURFACE ROUGHNESS NOT TO EXCEED 150 MICROINCHES RHR.
(b) TABULATED TORQUE VALUES INDICATE LIMITS FOR PROPER SEALING OF FITTING TO PAD.
(c) FOR STUDS AND CAP SCREWS ONLY.
(d) THREAD J APPLIES TO EXTERNAL PORTION OF STUDS AND INTERNAL TAPPED THREAD FOR CAP SCREWS.

CONNECTORS HAVE FIXED OR INTEGRAL MOUNTING FLANGES BUT ALL FITTINGS PROVIDED FOR A GIVEN TUBE OR HOSE SIZE ARE FULLY INTERCHANGEABLE IN MOUNTING DIMENSIONS.

MAX OPERATING PRESSURES: THRU 1 IN.- 1,500 PSI, 1-1/4 THRU 3 IN. - 500 PSI, ABOVE 3 IN. - 200 PSI.

MOUNTING WITH STUDS, STANDARD THRU-BOLTS WITH SELF-LOCKING NUTS OR CAP SCREWS WITH HEADS LOCKWIRED AND THREADED INTO HELICAL COIL OR THREADED INSERTS FOR SOFT METALS.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.005.

THIS IS A DESIGN STANDARD. NOT TO BE USED AS A PART NUMBER.

FED. SUP CLASS
4730

APPROVED 1 Apr 59 **REVISED** Ⓐ 1 FEB 85

P.A. AIR FORCE-82 Ⓐ

Other Cust
NAVY- AS
USAF- 99

TITLE

FITTING INSTALLATION, FLARED TUBE AND HOSE, SWIVEL.

MILITARY STANDARD

MS 33786

SHEET 1 **OF** 2

PROCUREMENT SPECIFICATION
NONE

SUPERSEDES:
AND10086

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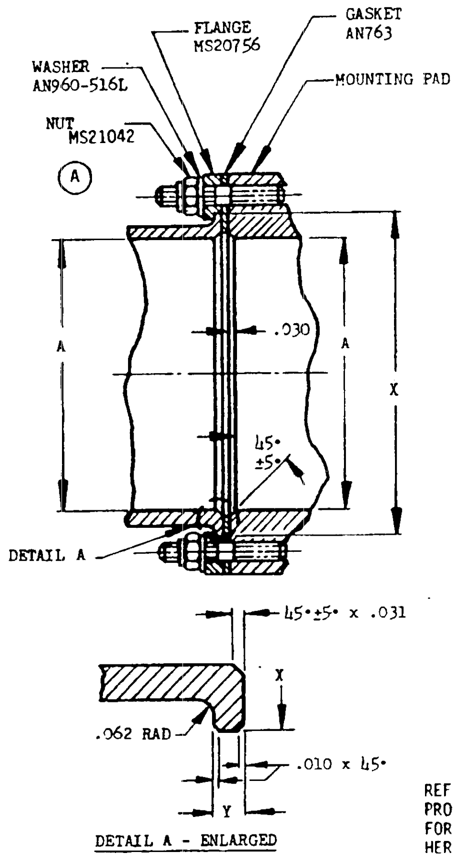
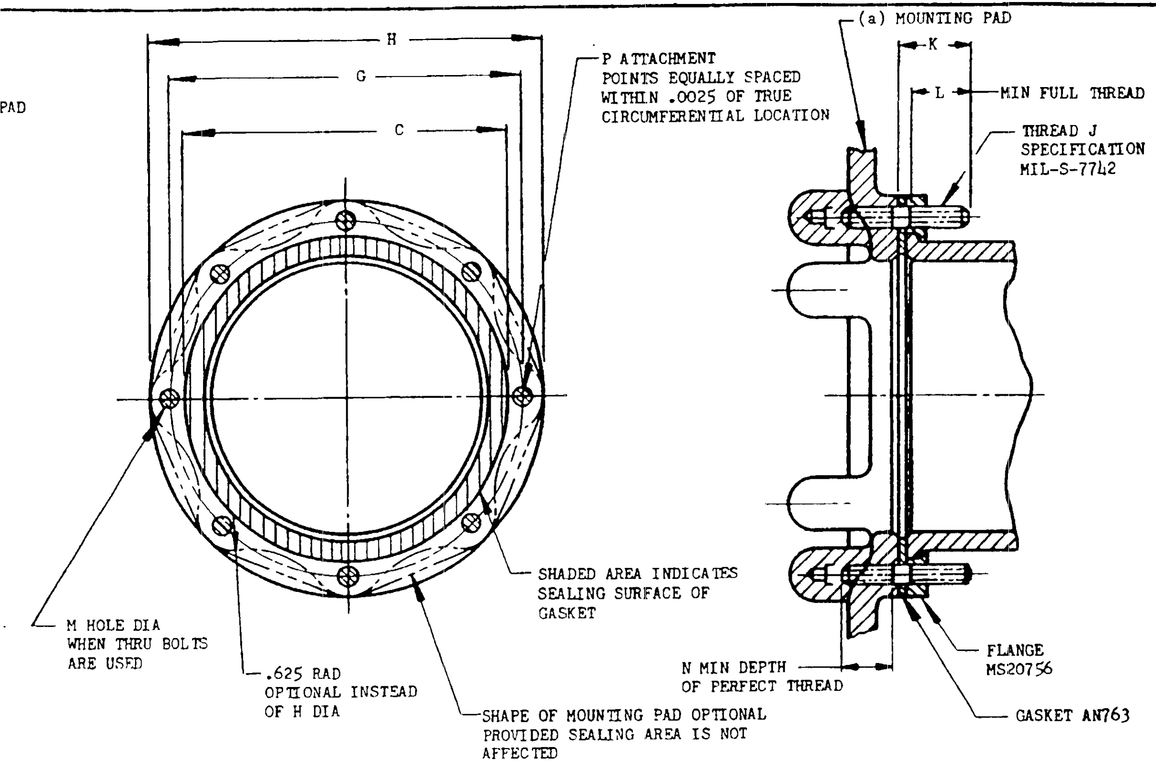
Review activities:
DLA-CS

User activities: ARMY - ME





REFERENCE DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BIDS OR REQUEST FOR PROPOSAL, EXCEPT THAT REFERENCED ADOPTED INDUSTRY DOCUMENTS SHALL GIVE THE DATE OF THE ISSUE ADOPTED. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.

DASH NO. REF	NOMINAL TUBE AND HOSE SIZE (REF)	A +.020 -.005 DIA	C DIA		G ±.005 DIA	H MIN DIA	THREAD J (d) UNF-3A	K ±.020	L	M BOLT HOLE +.010 -.000	(c) N	P	X		Y +.005 -.000	(b) WRENCH TORQUE INCH LBS.
56	3.500	3.281	4.281	+.020 -.000	4.875	5.562	5/16-24	.812	.688	.328	.469	8	4.214	+.000 -.010	.188	225
64	4.000	3.781	4.781		5.438	6.125							4.777			
80	5.000	4.781	5.781		6.438	7.125						10	5.777		.208	
96	6.000	5.781	6.781	+.030 -.000	7.438	8.125						12	6.777	+.000 -.020		
112	7.000	6.781	7.781		8.500	9.188						14	7.840		.220	

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.005, ANGLES ±1/2°.

SEE SHEET 1 FOR NOTES.

FED. SUP CLASS

APPROVED 1 APR 59 **REVISED** Ⓐ FOR CHANGES SEE SHEETS 1 AND 2

P.A. AIR FORCE - 82 Ⓐ **Other Cust** AIR FORCE -99 NAVY - AS	**TITLE** FITTING INSTALLATION, FLARED TUBE AND HOSE, SWIVEL.	**MILITARY STANDARD** **MS 33786**
PROCUREMENT SPECIFICATION NONE	**SUPERSEDES:** AND 10086	**SHEET** 2 **OF** 2

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